
A SPECIMEN OF CHAPTER THE FIRST

THE PRINCIPLES OF MANAGEMENT



A First Book

TO THE READER

This book is written for a reader who knows nothing about business and wishes to learn it from the beginning. You do not need earlier knowledge. Every word that may be new is explained in plain English when it first appears, and the book moves slowly where an idea is hard.

When a word appears for the first time, it is printed in **bold** type, and its meaning follows at once. The same meaning is repeated in the narrow column at the side of the page, in a short line beginning with the word in bold. A fuller list is kept in the glossary at the back of the book. After its first appearance, the word is printed in ordinary type.

Beside almost every paragraph, you will find one line in *italic* type. It states the paragraph's main idea. If you lose the thread, read these lines alone, from the top of the page to the bottom, and they will carry you through the argument.

Small side points appear in footnotes at the foot of the page.¹ At the end of each section, a short passage called *Before moving on* repeats the main ideas. Where an idea is difficult, a line called *You should now be able to say* gives you a sentence to test yourself against, and a *Pause and check* question follows, with its answer in the side column.

Throughout the book, we follow one made-up business, *Holloway & Sons Bakehouse*, to see each idea at work. It is *illustrative*: it is invented for teaching and is not a real company. Its facts are collected in tables, so that you can trace how it grows from chapter to chapter.

How this book is set out.

Bold type marks a new word, and the side column repeats its meaning.

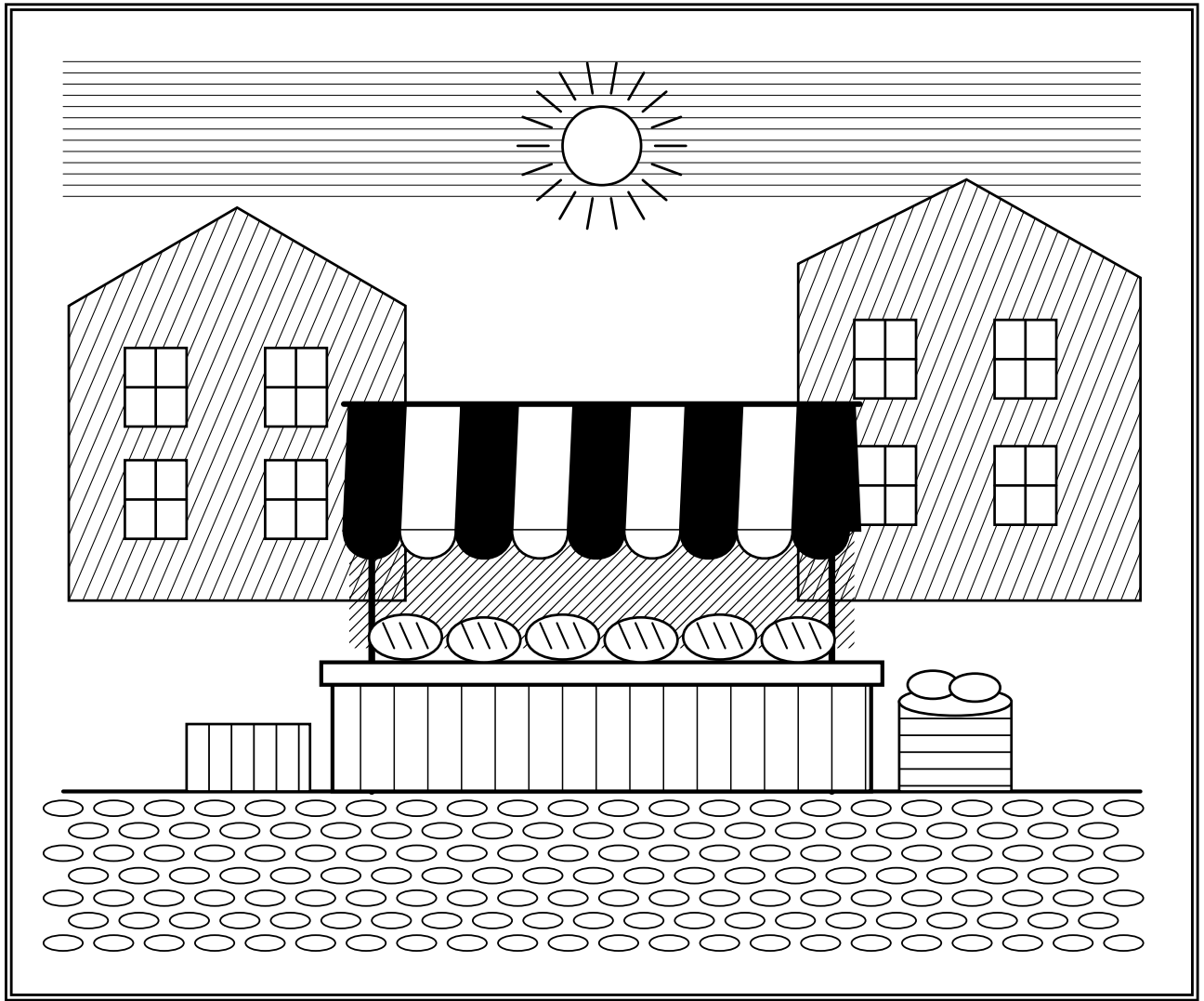
Key term. A new word, marked in bold where it first appears.

Italic lines in the side column summarise each paragraph.

Footnotes hold small points; recaps and self-tests close each part.

One illustrative business runs through the book.

1. A footnote looks like this. It adds a detail without breaking the main argument.



CHAPTER THE FIRST

THE IDEA OF MANAGEMENT

Who decides when three people must do one job?



A market stall, three tireless workers, and a day that goes wrong. From this small puzzle the chapter builds its first answers: what an organisation is, what management means, why it matters, and how to tell whether it is working.

A PUZZLE AT THE MARKET STALL

Three people work hard, yet the stall does badly.

The day produces waste, lost customers and confusion.

The puzzle: great effort, poor result. What was missing?

This chapter builds the answer slowly, starting with the word "organisation".

Some goals are too large for one person to reach alone.

Organisation. A group of people who work together to reach a goal.

Goal. Something a person or group wants to achieve.

Organisations come in many kinds, but each has people and a goal.

Working together gains speed, skill and safety, but creates new problems.

Task. A piece of work that someone must do.

Imagine a bread stall in a market square on a busy Saturday. Three people run it: two sisters, Anna and Bea, and their cousin Carl. All three arrive before dawn. All three work without rest until evening. Nobody is lazy, and nobody is careless. Yet at the end of the day the results are poor.

Forty loaves lie unsold on the shelf, and most of them are brown loaves. The white bread sold out by ten in the morning, and a line of customers walked away empty-handed. Anna and Bea both served at the counter during the quiet hours, while nobody was baking. At noon, Carl spent an hour searching for the coins needed to give change. By evening, all three were tired, and the stall had earned less than on an ordinary day.

Here is the puzzle. The effort was high. The people were capable. The bread was good. So why was the result poor? Something was missing, but it was not work, skill or bread. It was something harder to see. Before you read on, try to name it in your own words. Write down your answer; we shall return to it at the end of this chapter.

You will need. Nothing. This chapter begins from ordinary life, and it assumes no earlier knowledge of business. Many students find the first pages of a business book hard, because every word is new. Here, each new word is explained when it first appears, and it is marked in **bold** type. We begin with the most useful word of all: *organisation*.

1.1 PEOPLE WORKING TOGETHER

Why people form organisations

Think of a family that wants a house built. One person alone could lay a few bricks in a week. A group of ten, each with a clear task, could finish the walls in a month. The group can do what the single person cannot. This is the oldest reason for working together. An **organisation** means a group of people who work together to reach a goal. A **goal** means something a person or group wants to achieve. The bread stall is an organisation, and its goal is to sell bread at a fair profit.

Organisations are everywhere. A school is one, and so are a hospital, a football club, a bank and a factory. Some exist to earn money; others exist to teach, to heal or to serve the public. The goals differ, but the pattern is the same: several people, one shared goal, and work divided among them. In this book, we study the first kind most closely, the business, but the ideas also fit the others.

Working together brings three gains. First, work is done faster, because many hands share it. Second, each person can learn one **task** well. A task means a piece of work that someone must do, such as baking or serving. Third, risk is shared, because one person's illness does not stop the whole group. But working together also creates a new problem. Someone must decide who does which task,

and when. Without that decision, people collide, repeat each other's work, or leave gaps. This is exactly what happened at the stall.

What an organisation uses: resources

Pause for a moment and ask what the stall needed to sell bread. It needed flour and yeast, an oven, a table, coins for change, knowledge of how to bake, and the working hours of three people. Each of these is a **resource**. A resource means anything an organisation uses to reach its goal. A baker's flour is a resource; so is the baker's skill. The word is wider than most people expect, and it includes things you cannot touch.

It helps to sort resources into four groups. **Human resources** means the people who work in an organisation, with their skills and effort. *Financial resources* are the money available to pay for things. *Physical resources* are the materials, tools, buildings and machines. *Information resources* are the knowledge and facts needed to decide and act, such as how many loaves customers buy on a Saturday. In Chapter 2 we return to these four groups and see how managers choose among them.

The most important fact about resources is that they are limited. The stall has only one oven and only so many hours in a day. If Anna and Bea both serve at the counter, then nobody is baking, and the oven stands idle. Every hour of work and every kilogram of flour can be used in one way only, so using them badly is a real loss. Choosing how to use limited resources is the heart of what follows.

Before moving on. An organisation is people working together for a goal. Each person does a task. The organisation uses resources: people, money, materials and information. Resources are limited, so someone must decide how to use them.

You should now be able to say. "An organisation is a group of people who work together to reach a goal, and it uses limited resources to do so." Pause and check: name three resources that a school uses.

Everyone needs things to work with; we call them resources.

Resource. Anything an organisation uses to reach its goal.

Resources fall into four groups: people, money, materials and information.

Human resources. The people who work in an organisation, with their skills and effort.

Resources are limited, so using them well matters.

Recap of section 1.1.

A self-test for the reader.

Answer: Flour, oven, coins, knowledge, working hours.

1.2 DEFINING MANAGEMENT

The word and its early meaning

Every business student meets the word *management* in the first week, and most use it without being able to define it. The word has a humble origin. It comes from an old Italian verb meaning to handle or train a horse.¹ A rider does not pull a horse by force. The rider guides, corrects and directs its strength toward a place. That picture is helpful. A manager guides the strength of many people toward a goal.

The word "manage" comes from handling a horse.

A landmark definition

In 1916, Henri Fayol described management by its activities.

1. The Italian verb is *maneggiare*, from *mano*, the word for "hand". The idea of guiding something powerful with a steady hand has stayed in the word.

Many writers have defined management, and their definitions differ in wording. One of the most influential comes from Henri Fayol (1841–1925), a French mining engineer who led a large mining company for thirty years.² In 1916, he published a book on the subject. He said that to manage is to plan, to organise, to command, to coordinate and to control. He did not offer a short sentence. He offered a list of activities, because he believed a manager is known by what a manager does.

Fayol's list shows management as a set of connected activities.

Fayol's list has lasted for more than a century, although later writers shortened it and changed some words. It teaches two lessons. First, management is not one act but several, and they depend on each other. A plan without control cannot be checked; control without a plan has nothing to check against. Second, management is a matter of *doing*. A person does not become a manager by holding a title, but by performing these activities for others. We study each activity in its own chapter later.

Building a definition word by word

We build a plain definition from four parts.

Management. The process of working with people and resources to reach goals efficiently and effectively.

Manager. A person who is responsible for the work of other people.

Management is a process, not a single act.

Let us now build a definition that a beginner can hold in one sentence. **Management** means the process of working with people and resources to reach goals efficiently and effectively. A **manager** means a person who is responsible for the work of other people. Four parts of the definition need separate attention: *process, people and resources, goals, and efficiently and effectively*. We take them in turn.

The word *process* means a series of actions that follow one another and continue over time. A manager does not plan on Monday and finish the job. The manager plans, organises, leads and checks again and again, because every week brings new conditions. Think of a ship's captain, who checks the compass again and again during the whole voyage. So management is never finished; it only moves from one round to the next.

The second part says that managers work *with* people and resources. A manager who bakes every loaf personally is a baker, not a manager. A manager gets results through the work of others, and so the manager must understand people: what they can do, what they need and why they work. The resources of the previous section, such as flour, money and information, are also in the manager's care. But the human resources require the most care, because people can decide, refuse, learn and care.

The third part says that management aims at *goals*. Work with no goal is only activity. The stall at the start of this chapter had effort, but it had no clear goal for the day, such as "sell three hundred loaves, half white and half brown". Without that goal, nobody could tell whether Saturday had gone well or badly. A goal tells everyone what they are working toward, and it lets the manager check the result.

Managers work with people and resources, and the people matter most.

Management always serves a goal.

2. Fayol studied at the mining school at Saint-Etienne and joined a mining and iron company in 1860. He became its managing director in 1888 and held the post until 1918.

"Efficiently and effectively" sets the standard for good management.

The fourth part sets the standard. Reaching a goal is not enough; it must be reached with care for the resources used (*efficiently*) and in a way that truly achieves what was wanted (*effectively*). We return to these two words in section 1.5, where they get a full explanation. For now, remember only that both are needed. A stall that sells every loaf but wastes half its flour has reached the goal badly.

Three seats, one idea

Ideas in business look different from different seats. The *owner* of the stall sees management as protection for the money invested. A *worker* sees it as a source of clear instructions, fair treatment and safe conditions. A *customer* sees it only through its results: fresh bread, a fair price and a short line. A good manager keeps all three seats in mind, because a decision that pleases one and harms the others will not last.

The same idea looks different to an owner, a worker and a customer.

Where the definition is tested

Every definition has edges, and a careful student should find them. First, ownership and management are different things. An owner supplies money and takes the risk of loss; a manager directs the work. One person may do both, as at a small stall, but in a large company the owners are thousands of shareholders who never meet the workers. Second, management is not the same as administration in the narrow sense of keeping records. Records help a manager; they are not the whole task.

Not every owner is a manager, and not every manager is an owner.

Third, the word *manager* is not limited to people with that title on a door. A senior nurse who plans the ward's work is managing. A team member who organises a school project is managing. Even a single person managing a personal budget uses the same activities on a small scale. This wider view explains why management ideas are useful far beyond business.

People also manage without the title; self-management is one example.

Before moving on. Management is the process of working with people and resources to reach goals efficiently and effectively. A manager is responsible for the work of others. Fayol listed the activities as planning, organising, commanding, coordinating and controlling. Ownership and management are different things.

Recap of section 1.2.

You should now be able to say. "Management is the process of working with people and resources to reach goals efficiently and effectively." Pause and check: at a stall run by one person, who is the owner and who is the manager?

A self-test for the reader.

Answer: The owner may supply money; the manager directs the work.

1.3 THE NATURE AND PURPOSE OF MANAGEMENT

What management is like: its nature

By the *nature* of management we mean its main features: what kind of activity it is. Six features appear in almost every description. We list them first, and then look at each one. Management is *goal-directed*, *universal*, *a process of several activities*, *carried out at different levels*, *concerned with people*, and *partly exact, partly personal*.

The nature of a subject means its main features.

Management is goal-directed,

The first feature is that management is goal-directed. Everything a manager does is judged by whether it moves the organisation toward its goals. A manager may be kind, clever and well liked, but if the goals are missed, the management has failed. This feature links the chapter's first three ideas: people, resources and goals form one chain.

Management is universal: every organisation needs it.

Management is a continuing process with linked activities.

Management happens at many levels of an organisation.

Management deals with people, who are different from machines.

Management mixes exact knowledge with personal skill.

Purpose means the reason management exists.

The main purpose is to reach the organisation's goals.

Management also balances interests and helps the organisation cope with change.

which means it always aims at a result.

The second feature is that management is universal. A bakery, a hospital, a government office and a village club all face the same tasks: set goals, arrange work, guide people and check results. The details change from place to place, but the activities remain. This is why a student who learns the principles of management in one setting can apply them in another, with care for differences.

The third feature is that management is a process of several linked activities: planning, organising, leading and controlling. They are not separate jobs done by separate people. One manager may move between them in a single hour. We keep them apart in this book only because it is easier to learn them one at a time.

The fourth feature is that management occurs at several levels. In a small stall, one person may manage everything. In a large bakery, there may be a director, several department managers and team supervisors. Each level manages a different part of the whole, and each needs somewhat different skills. Chapter 2 explains the levels in detail.

The fifth feature is that management is concerned with people. Machines do what they are set to do. People respond differently from day to day, each with hopes, worries and habits. This makes management harder than operating a machine, and it explains why a large part of this book is about people: their motives, their groups, and their leaders.

The sixth feature needs care. Management has organised knowledge that can be learned and tested, and this gives it some of the character of a science. But applying that knowledge to a real person in a real situation requires judgement and skill, and this gives it the character of an art. Section 1.6 returns to this point.

What management is for: its purpose

The *purpose* of management is the reason it exists: what it is meant to achieve. Nature describes what management is like; purpose explains why it matters. Think of a bridge. Its nature is steel, stone and shape. Its purpose is to carry people safely across a river. Both descriptions are true, and each answers a different question.

The first purpose is to help the organisation reach its goals. Good management turns the scattered effort of many people into one directed effort. At the market stall, the effort was large but scattered, and so the result was poor. The second purpose follows from the first: to use resources with little waste, since resources are limited, as we saw in section 1.1.

The third purpose is to balance the interests of different groups. Owners want profit; workers want fair pay; customers want low prices and good quality; society wants honest and safe business. These wants often pull in different directions, and the manager must

find a balance that all can accept. The fourth purpose is to help the organisation meet change. Prices rise, tastes shift, new rules appear, and a manager must adjust the plan without losing the goal.

A fifth purpose joins steadiness and growth. Steadiness means that the organisation keeps working from day to day, with few shocks. Growth means it gains more customers, skills and resources over time. Good management provides both: a steady base today and a wider base tomorrow.

A further purpose: keep the organisation steady and help it grow.

Before moving on. The nature of management is its main features: it is goal-directed, universal, a process, found at many levels, concerned with people, and partly a science and partly an art. The purpose of management is to reach goals, use resources well, balance interests, meet change, and keep the organisation steady and growing.

Recap of section 1.3.

You should now be able to say. "The nature of management is what it is like; its purpose is what it is for." Pause and check: in your own words, what is the difference between nature and purpose?

A self-test for the reader.

Answer: Nature describes what it is like; purpose states why it matters.

1.4 WHY MANAGEMENT MATTERS

Why does a business student need to study management at all? The answer is that nothing in an organisation runs itself. Resources do not arrange themselves, and goals do not reach themselves. Management is the force that turns separate parts into a working whole, and its importance shows in six ways.

Without management, effort is scattered; with it, effort is directed.

First, management gives direction. It tells people where the organisation is going, and it joins their separate efforts into one. Second, it makes better use of resources. By planning who does what and when, a manager reduces waste, such as idle ovens and unsold bread. Third, it creates order. Clear tasks and clear reporting lines reduce quarrels and repeated work.

Management gives direction and joins effort.

Fourth, management helps the organisation adjust to change. A manager watches prices, customers and rules, and alters plans in good time. Fifth, it supports growth: an organisation that is run well can add customers, workers and skills without falling into disorder. Sixth, it serves society. Well-run organisations provide jobs, goods and services, and they pay for them honestly. Poor management wastes the resources of a whole community.

Management also helps organisations adjust and grow.

A careful reader will notice a limit. Management cannot rescue a product nobody wants, and it cannot replace honesty or skill. A well-managed bakery that bakes bad bread will still fail. Management makes good products and good people more effective; it does not replace them.

Importance has limits: management cannot replace good products or honest work.

1.5 PRODUCTIVITY, EFFICIENCY AND EFFECTIVENESS

Go slowly here.

Three words that are often mixed up need slow, careful treatment.

Three words appear again and again in the study of management: *productivity*, *efficiency* and *effectiveness*. They sound alike, and many students use them as if they meant the same thing. They do not. Each answers a different question about how well an organisation is working. We take them one at a time, using small numbers first.

Everyday bridge: how many sandwiches in an hour?

Productivity compares what you produce with what you use.

Output. What an organisation produces, such as loaves of bread.

Input. What an organisation uses to produce, such as hours of work or kilograms of flour.

Productivity. The amount of output produced from each unit of input.

Labour hour. One hour of work by one person.

First worked example, with small clean numbers.

Second worked example, with messier and more realistic numbers.

Common slips: dividing the wrong way, mixing periods, forgetting units.

Recap of productivity.

Productivity: how much for how much?

Start with an everyday case. Suppose you make sandwiches for a school event. In one hour you make twenty. In the next hour, with a better method, you make twenty-five. You are now more productive, because you get more sandwiches from the same hour of work. Nothing else has changed. This is the idea of productivity.

Output means what an organisation produces, such as loaves of bread. **Input** means what an organisation uses to produce, such as hours of work or kilograms of flour. **Productivity** means the amount of output produced from each unit of input. A **labour hour** means one hour of work by one person. If two people each work three hours, that is six labour hours. In words, productivity is output divided by input. In symbols, $P = O \div I$, where P is productivity, O is output and I is input.

Given. A stall bakes 60 loaves using 4 labour hours.

Find. The productivity of the stall, in loaves per labour hour.

Formula. Productivity = output $\div$ input, that is, $P = O \div I$.

Steps. $P = 60 \text{ loaves} \div 4 \text{ labour hours} = 15 \text{ loaves per labour hour}$.

Answer. The stall produces 15 loaves for each labour hour.

Check. Multiply back: $15 \text{ loaves per hour} \times 4 \text{ hours} = 60 \text{ loaves}$. The check agrees with the given output.

Given. In Week 1, a bakery produces 450 loaves using 90 labour hours. In Week 2, it produces 520 loaves using 100 labour hours.

Find. The change in productivity from Week 1 to Week 2, as a percentage.

Formula. Percentage change = $(\text{new value} - \text{old value}) \div \text{old value} \times 100$.

Steps. Week 1: $450 \div 90 = 5.0 \text{ loaves per labour hour}$. Week 2: $520 \div 100 = 5.2 \text{ loaves per labour hour}$. Change = $(5.2 - 5.0) \div 5.0 \times 100 = 0.2 \div 5.0 \times 100 = 4 \text{ per cent}$.

Answer. Productivity rose by 4 per cent, even though the bakery used more hours.

Check. Increase the old value by 4 per cent: $5.0 \times 1.04 = 5.2$, which equals the new value.

Three slips appear often. The first is dividing input by output, which gives hours per loaf, not loaves per hour. Both are valid measures, but they say different things, so state which one you use. The second is mixing periods, such as comparing loaves per day with loaves per week. The third is forgetting the units: a bare number like “15” has no meaning until you write “loaves per labour hour”. When finding a percentage change, always divide by the *old* value.

Before moving on. Productivity is output divided by input. It is a number, and it can be compared from week to week or between two organisations. Raising it means getting more output from the same input, or the same output from less input.

Efficiency: how little waste?

Now consider a different question. Two people each cook the same meal. One uses all of the vegetables; the other throws away a third of them. Both cooks make dinner, but the first wasted less. This is the idea behind efficiency.

Efficiency means using resources with as little waste as possible to reach a result. To measure it, we compare the resources a task *should* need with the resources it actually used. In words, efficiency is the resources needed divided by the resources used, times one hundred. In symbols, $E = (R \text{ needed} \div R \text{ used}) \times 100$ per cent, where E is efficiency and R stands for the resource being measured. A result of 100 per cent means no waste at all.

Given. Baking 100 loaves should need 25 kilograms of flour. A stall used 30 kilograms.

Find. The efficiency of flour use.

Formula. Efficiency = (resource needed $\div$ resource used) $\times$ 100.

Steps. $E = (25 \div 30) \times 100 = 0.8333 \times 100 = 83.3$ per cent.

Answer. The stall used flour with 83.3 per cent efficiency; the rest was waste.

Check. Waste is $30 - 25 = 5$ kilograms. As a share of the 30 kilograms used, that is $5 \div 30 = 16.7$ per cent. And $100 - 16.7 = 83.3$, so the answer agrees.

A reader may ask how efficiency differs from productivity, since both compare output with input. The difference lies in what is compared with what. Productivity states a rate: so many loaves per hour. Efficiency judges that rate against what is possible: how much was wasted. In practice, cutting waste usually raises productivity, so the two often move together.

Effectiveness: was the goal reached?

Last, consider a student who spends the whole weekend making a beautiful poster. The poster was made without waste and with great speed. But the teacher asked for a written report. The poster is useless for the task. The student was efficient but not effective. This is the idea of effectiveness.

Effectiveness means the extent to which an organisation reaches its goals. It does not ask how much was used or wasted; it asks only whether the goal was reached, and how far. In words, effectiveness is the result achieved divided by the result wanted, times one hundred. Suppose the goal is to sell 500 loaves and the stall sells 450. Then effectiveness is $(450 \div 500) \times 100 = 90$ per cent. Check: 90 per cent of 500 is 450.

Everyday bridge: cooking with and without waste.

Efficiency compares resources used with resources that should be needed.

Efficiency. Using resources with as little waste as possible to reach a result.

A worked example of efficiency.

Productivity and efficiency are related but not the same.

Everyday bridge: finishing the wrong job very well.

Effectiveness asks whether the goal was reached.

Effectiveness. The extent to which an organisation reaches its goals.

Many people say efficiency is “doing things right” and effectiveness “doing the right things”.

People often summarise the difference in a short saying: efficiency is doing things right, and effectiveness is doing the right things. The saying is useful as a memory aid. But the measures above give it exact form: efficiency looks at waste, and effectiveness looks at the goal.

The four possible cases show why both are needed.

A table helps here. Read it by choosing one row at a time and asking: was the waste low, and was the goal reached? The stall in each row bakes bread for a goal of 500 loaves sold.

Table 1. The four cases of efficiency and effectiveness

Case	What happens	Result
Efficient and effective	The goal is reached with little waste	Best result
Efficient but not effective	Little waste, but the goal is missed (the wrong loaves are baked)	Resources are spent well on the wrong thing
Effective but not efficient	The goal is reached, but with much waste	Success at a high price
Neither	The goal is missed and resources are wasted	Worst result

The three ideas compared in one table.

We can now set the three ideas side by side. Notice that each row asks a different question, and that each needs a different measure. Read across a row to see how the three ideas differ on one point.

Table 2. Productivity, efficiency and effectiveness compared

Point	Productivity	Efficiency	Effectiveness
Question asked	How much output from each unit of input?	How little was wasted?	Was the goal reached?
How measured	Output ÷ input	(Needed ÷ used) × 100	(Achieved ÷ wanted) × 100
Bakery example	15 loaves per labour hour	83.3 per cent of flour used well	90 per cent of the sales goal met

Common confusion: high productivity does not prove effectiveness.

Many students find this step confusing, because high productivity looks like success. But a bakery can bake 20 loaves per hour of a bread nobody buys. Its productivity is high, its efficiency may be high, and its effectiveness is zero. A manager must therefore watch all three, and must never trust one number alone.

Before moving on. Productivity is output per unit of input. Efficiency is how little waste there is. Effectiveness is how far the goal was reached. A good manager aims at all three. Pause and check: a stall bakes 200 loaves but sells only 150 of a goal of 200. Is the problem mainly one of effectiveness or of efficiency?

You should now be able to say. “Productivity counts output per input, efficiency counts waste, and effectiveness counts goals reached.”

Recap of section 1.5.

Answer: Mainly effectiveness: only 75 per cent of the goal was reached, and 50 loaves were wasted.

1.6 MANAGEMENT AS A SCIENCE, AN ART AND A PROFESSION

Three descriptions of management are in common use.

Writers describe management in three ways: as a science, as an art and as a profession. Each description is partly right. To judge them, we need a clear meaning for each word, and then a test of how well management fits it.

Is management a science?

Science means organised knowledge, built from careful observation and tested by evidence. Physics is a clear example: its laws hold in the same way every time. Management has some of this character. There is a body of principles, such as the division of work, that managers have studied and tested over many years. A student can learn them from books.

There is a limit, however. Management deals with people, and people do not behave in the same way every time. A principle that works well in one bakery may fail in another, since the workers, customers and times differ. For this reason, management is called an *inexact* or *social* science. Its principles are guides, not fixed laws.

Is management an art?

Art means skill in applying knowledge to real situations, improved by practice. A person can read every book on swimming and still sink. Skill comes from doing it. Management is also a matter of doing: a manager must read a situation, choose among imperfect options and act, often with little time. Two managers with the same knowledge may get different results because their skill differs.

Hence the usual conclusion: management is both a science and an art. Science supplies the principles, and art supplies the skill to use them well. A doctor is a good example. A doctor studies the science of medicine, but also needs the art of dealing with each patient. Neither alone is enough.

Is management a profession?

Profession means an occupation with its own body of knowledge, trained members and standards of conduct. Medicine and law are the usual examples. Five features mark a profession. It has a body of knowledge. It requires formal training. Its members follow a code of conduct. Its members serve others, not only themselves. And, often, a professional body controls who may practise.

Test management against the five features. It has a growing body of knowledge. Many managers are trained in universities. Professional bodies for managers exist, and they publish codes of conduct. But a person can become a manager without formal training and without any licence, and no single body controls the entry. So management is moving toward the character of a profession, but it is not yet a full profession in the way that medicine is.

A science is organised knowledge tested by evidence.

Science. Organised knowledge, built from careful observation and tested by evidence.

But management is not an exact science.

An art is skill in using knowledge in real situations.

Art. Skill in applying knowledge to real situations, improved by practice.

Word in use. *Art.* In daily life, painting, music and drawing. In this section, personal skill in doing something well, as in "the art of cooking".

Management needs both knowledge and skill.

A profession has a body of knowledge, training, standards and a duty to serve.

Profession. An occupation with its own body of knowledge, trained members and standards of conduct.

Management meets some of these tests but not all.

Table 3. Management tested against the five features of a profession

Feature	Present in management?
Body of knowledge	Yes, and growing
Formal training	Common, but not required
Code of conduct	Exists in professional bodies, but not binding on all managers
Service to others	Expected, but not enforced
Control of entry	No

The three views reach one balanced conclusion.

Recap of section 1.6.

Answer: Both; science gives the principles, art the skill.

Putting the three views together gives a balanced answer. Management rests on organised knowledge (science), depends on personal skill (art), and is moving toward professional standing (profession). A student who says it is *only* one of these has missed something.

Before moving on. Science is tested, organised knowledge. Art is skill in applying it. A profession has knowledge, training, a code of conduct and a duty to serve. Management is both science and art, and is moving toward a profession.

You should now be able to say. “Management is partly a science, partly an art, and not yet a full profession.” Pause and check: why does management need art as well as science?

A WORKED CASE: HOLLOWAY & SONS BAKEHOUSE

From here, the book follows one illustrative business.

Throughout this book, we follow one business to see each idea at work. It is called *Holloway & Sons Bakehouse*. It is *illustrative*: it is invented for teaching, and it is not a real company. We record its facts in a case file, with dates and figures, and each chapter adds to the file. We begin with its first two weeks.

Table 4. Case file: Holloway & Sons Bakehouse, Year 1 (illustrative)

Item	Fact
Opened	March, Year 1, in rented premises in a market town
People	Edmund Holloway (owner and head baker); Tom Holloway, his son (sales and delivery); Ruth Pell (baker, hired)
Starting funds	2,400 currency units
Weekly goal	Sell 600 loaves a week
Weekly labour	120 labour hours (3 people × 8 hours × 5 days)
Flour standard	0.25 kg per loaf, so 150 kg for 600 loaves

The first week's records show a gap between goal and result.

Week 1 gave these records. The bakehouse baked 600 loaves and sold 540; the other 60 were thrown away. It used 120 labour hours and 165 kilograms of flour. Week 2 gave the following records. After Edmund made changes, it baked 570 loaves and sold 555. It used 120 labour hours and 152 kilograms of flour.

Situation. In Week 1, all three people worked the whole week, yet 60 loaves were wasted and the sales goal was missed. In Week 2, Edmund began to record sales by day and by kind of loaf. He baked according to the record, placed Tom at the counter in the busy morning hours, and kept Ruth at the oven. He also bought flour by weight and measured the amount used.

Question. Did the bakehouse improve between Week 1 and Week 2, and how do the three measures show it?

Reasoning. For output, we count loaves *sold*, since waste earns nothing. Week 1 productivity is $540 \div 120 = 4.5$ loaves per labour hour. Week 2 productivity is $555 \div 120 = 4.625$, or 4.63 rounded. The change is $(4.625 - 4.5) \div 4.5 \times 100 = 2.8$ per cent. Effectiveness against the goal of 600 is $540 \div 600 = 90$ per cent in Week 1, and $555 \div 600 = 92.5$ per cent in Week 2. For flour, 600 loaves should need 150 kilograms, so Week 1 efficiency is $150 \div 165 \times 100 = 90.9$ per cent. In Week 2, 570 loaves should need 142.5 kilograms, so efficiency is $142.5 \div 152 \times 100 = 93.8$ per cent.

Conclusion. All three measures improved, and none of the changes needed more people or more hours. Edmund set a goal, divided the tasks, directed each person, and checked the results. These are management activities. Even so, the bakehouse still sold 45 loaves fewer than its goal, so the work is not finished.

Check. Loaves sold never exceed loaves baked ($540 \leq 600$ and $555 \leq 570$). Waste fell from 60 to 15 loaves. Hours stayed at 120, so the gain in productivity came from better use of the same time.

We can now answer the puzzle from the start of the chapter. The stall at the market had hard-working people, good bread and a busy square. What it lacked was management. Nobody had set a goal for the day, such as how many loaves of each kind to bake. Nobody had divided the tasks, so two people served while no one baked. Nobody had checked results during the morning, so the white bread ran out unseen. And nobody had directed the effort when the coins went missing. Any of the three could have done these things, and the stall would have needed no fourth person.

Back to the puzzle: the Stall lacked management.



CLOSING PACK

Summary

People form organisations to reach goals that one person cannot reach alone. Organisations use limited resources, and so someone must decide how to use them. Management is the process of working with people and resources to reach goals efficiently and effectively. Its nature is goal-directed, universal, a process, found at many levels, concerned with people, and part science and part art. Its purpose is to reach goals, use resources well, balance interests, meet change, and keep the organisation steady and growing. Productivity, efficiency and effectiveness are three different measures. Management is both science and art, and is moving toward a profession.

Chapter 1 in brief.

A short case for the reader to work on.

Think it through

A school canteen is run by three staff. It serves 200 lunches each day, but about 30 meals are thrown away, and the line for the counter is long at noon. Name four resources of the canteen. State a possible goal for the canteen. Say which management activities are missing. Decide whether the main problem is one of efficiency, effectiveness or both, and give your reasons.

Five common misunderstandings, stated and corrected.

Where people go wrong

"A manager is a boss who gives orders." Giving orders is a small part of the work. A manager sets goals, arranges work, guides people and checks results. *"Efficiency and effectiveness mean the same."* They do not. One measures waste; the other measures whether the goal was reached. *"Management belongs only to business."* Every organisation needs it, including schools, hospitals and clubs. *"Hard work is enough."* The market stall shows that hard work without direction gives a poor result. *"Management must be either a science or an art."* It is both; it rests on knowledge, and it needs skill.

Robert Owen managed a cotton mill and treated its workers as people.

In history

Robert Owen (1771–1858), a Welsh-born businessman, took charge of the New Lanark cotton mills in Scotland in 1800. Many mills of that period employed very young children for long hours. Owen stopped taking the youngest children into the mills, and he built a school for the children of the village. In 1816, he opened a building called the Institution for the Formation of Character. He believed that fair treatment of workers was good for the workers and good for the business, and the mills continued to make a profit. His story is an early example of a manager who looked beyond machines to the people who worked them.

Where to read more.

Sources

Henri Fayol, *Administration industrielle et générale* (1916); English translation by Constance Storrs, *General and Industrial Management* (1949). This book first set out the activities of management. Robert Owen, *A New View of Society* (1813–1816), which describes his ideas about character and work.



Questions for the reader; short answers appear at the back.

REVIEW QUESTIONS

1. Define an organisation and give one example.
2. Name the four groups of resources, with one example of each.
3. Define management in one sentence.
4. State one difference between an owner and a manager.
5. List the six features of the nature of management.
6. State three purposes of management.
7. Distinguish productivity, efficiency and effectiveness.

8. Explain why management is called both a science and an art.
9. Is management a profession? Give reasons.

Practice Problems

1. A stall bakes 96 rolls using 8 labour hours. Find its productivity.
2. In Week 3, the bakehouse sold 570 loaves in 120 labour hours. Find the productivity, and the percentage change from Week 2 (4.625 loaves per labour hour).
3. A goal is to sell 600 loaves, and 510 are sold. Find the effectiveness.
4. Baking should need 120 kilograms of flour, but 150 kilograms were used. Find the efficiency of flour use.

Calculations for the reader; follow the order Given, Find, Formula, Steps, Answer, Check.

ANSWERS TO REVIEW QUESTIONS AND PRACTICE PROBLEMS

Review.

1. A group of people working together to reach a goal; for example, a school.
2. People (workers), money (cash), physical (ovens, flour), information (sales records).
3. Management is the process of working with people and resources to reach goals efficiently and effectively.
4. An owner supplies money and takes the risk; a manager directs the work.
5. Goal-directed, universal, a process, found at many levels, concerned with people, part science and part art.
6. Any three: reach goals, use resources well, balance interests, meet change, keep steady and growing.
7. Productivity is output per input; efficiency is how little is wasted; effectiveness is how far the goal was reached.
8. Science supplies tested principles; art supplies the skill to apply them to people.
9. Partly: it has knowledge and training, but no required licence or single controlling body.

Short answers.

Practice.

1. $96 \div 8 = 12$ rolls per labour hour. Check: $12 \times 8 = 96$.
2. $570 \div 120 = 4.75$ loaves per labour hour. Change = $(4.75 - 4.625) \div 4.625 \times 100 = 0.125 \div 4.625 \times 100 = 2.7$ per cent. Check: $4.625 \times 1.027 = 4.75$.
3. $510 \div 600 \times 100 = 85$ per cent. Check: 85 per cent of 600 is 510.
4. $120 \div 150 \times 100 = 80$ per cent. Check: waste is 30 kg, which is 20 per cent of 150.

Answers to the practice problems.